

Assessment Tool CO1 Concept Mapping

Q1. Complete the OSI Concept Map

OSI Model:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Missing Layers: Session Layer, Data Link Layer.

Functions:

- Session Layer: Controls communication, manages sessions, and restores communication if interrupted.
- Data Link Layer: Organizes data into frames, provides MAC addressing, detects errors, and controls data flow.

Q2. Data Units and OSI Layers

Application/Presentation/Session → Data

Transport → Segment

Network → Packet

Data Link → Frame

Physical → Bits

Transformation: Data → Segment → Packet → Frame → Bits.

Analysis: Encapsulation allows reliable and accurate transmission by adding information at each layer.

Q3. MAC Address and IP Address

Data Link Layer → MAC Address → Local Communication.

Network Layer → IP Address → Communication Across Networks.

Analysis: MAC identifies devices in a LAN, while IP identifies devices across different networks.

Q4. Error Control

Data Link Layer: Detects errors using CRC or checksum.

Transport Layer: Ensures reliable delivery using acknowledgements and retransmission.

Analysis: Together they improve communication reliability.

Q5. Web Browsing Using OSI Layers

User Opens Browser → Application → Presentation → Session → Transport → Network → Data Link → Physical → Internet → Web Server Response.

Failure Analysis:

- Application: Website request fails.
- Presentation: Encryption issues.
- Session: Session interrupted.
- Transport: Data transfer fails.
- Network: Routing problems.
- Data Link: Frame errors.
- Physical: Cable/Wi-Fi failure.

Analysis: Every OSI layer has a specific role; failure in any layer interrupts communication.